BASICS OF QUANTUM COMPUTING

- $|0\rangle := \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle := \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

- $|+\rangle := \frac{|0\rangle + |1\rangle}{\sqrt{2}}$; $|-\rangle := \frac{|0\rangle - |1\rangle}{\sqrt{2}}$; $|\pm_Y\rangle = \frac{|0\rangle \pm i|1\rangle}{\sqrt{2}}$

- X , Y , Z Pauli : 

- P Pauli $\Rightarrow tr(P) = 0$, $P = P^+$, $P^2 = \mathbb{1}$. $tr(XY) = 0$, $tr(YZ) = 0$, $tr(XZ) = 0$

- $X|+\rangle = +|+\rangle$, $X|-\rangle = -|-\rangle$;
  $Y|\pm_Y\rangle = \pm|\pm_Y\rangle$

- $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$ $\Rightarrow H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $H^2 = \mathbb{1}$, $H^+ = H$.

- $Z|+\rangle = |-\rangle$, $Z|-\rangle = |+\rangle$

- If $\mathcal{H} = (\mathbb{C}^2)^{\otimes h}$, $h=2 \Rightarrow$

$$|0\rangle_{1,2} = |0\rangle_1 \otimes |0\rangle_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$|1\rangle_{1,2} = |0\rangle_1 \otimes |1\rangle_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1\cdot\begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0\cdot\begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$|2\rangle_{1,2} = |1\rangle_1 \otimes |0\rangle_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$|3\rangle_{1,2} = |1\rangle_1 \otimes |1\rangle_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

- If $\mathcal{H} = (\mathbb{C}^2)^{\otimes h} \Rightarrow$

$$|0\rangle := |0\rangle_1 \otimes \cdots \otimes |0\rangle_n = |0\rangle^{\otimes N}$$

$$|1\rangle := |0\rangle_1 \otimes \cdots \otimes |1\rangle_n ;$$

$$|2\rangle := |0\rangle_1 \otimes \cdots \otimes |1\rangle_{n-1} \otimes |0\rangle_n ;$$

$$\vdots$$

$$|2^n - 1\rangle := |1\rangle_1 \otimes \cdots \otimes |1\rangle_n$$

- $H^{\otimes N}|0\rangle^{\otimes N} = (H|0\rangle) \otimes (H|0\rangle) \otimes \cdots \otimes (H|0\rangle) = |+\rangle \otimes \cdots \otimes |+\rangle = |+\rangle^{\otimes N}$

- $|+\rangle^{\otimes N} = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes \cdots \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) = \frac{1}{\sqrt{2^n}} \sum_{x_1, \ldots, x_n \in \{0,1\}^n} \overbrace{|x_1, \ldots, x_n\rangle}^{|x_1\rangle \otimes \cdots \otimes |x_n\rangle}$

- $|0\rangle$ — H
  $|0\rangle$ — H
  $|0\rangle$ — H
  $= H^{\otimes N} |0\rangle^{\otimes N} = |+\rangle^{\otimes N}$

NOT ENTANGLED!

- if $x = 0, 1 \Rightarrow H|x\rangle = \frac{|0\rangle + (-1)^x |1\rangle}{\sqrt{2}} = \sum_{z=\{0,1\}} \frac{(-1)^{x \cdot z}}{\sqrt{2}} |z\rangle$

- $H^{\otimes n} |x_1, \dots, x_n\rangle = \sum_{z_1=\{0,1\}} \frac{(-1)^{x_1 z_1}}{\sqrt{2}} |z_1\rangle \otimes \dots \otimes \sum_{z_n=\{0,1\}} \frac{(-1)^{x_n z_n}}{\sqrt{2}} |z_n\rangle$

$$= \frac{1}{\sqrt{2^n}} \sum_{z_1, \dots, z_n = \{0,1\}} (-1)^{x_1 z_1 + \dots + x_n z_n} |z_1, \dots, z_n\rangle$$

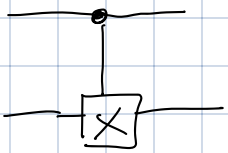
$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{x \cdot z} |z\rangle$ with $x \cdot z = x_1 z_1 + \dots + x_n z_n$.

PAULI STRINGS

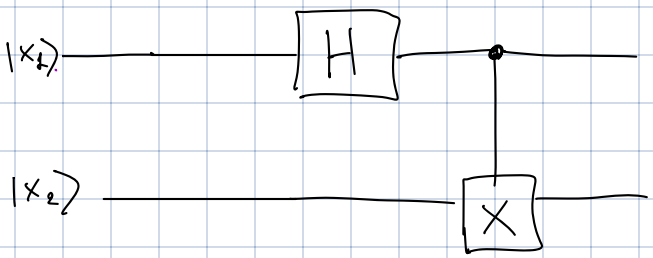
- $P = P_1 \otimes \dots \otimes P_n$ with $P_i = \{\mathbb{1}, X, Y, Z\} \Rightarrow \mathrm{tr}(P) = 0$, $P = P^\dagger$, $P^2 = \mathbb{1}$.

- P^i, P^s Pauli strings $\Rightarrow \mathrm{tr}(P_i P_s) = 2^n \delta_{i,s}$.

- $CNOT_{(A,B)} := |0\rangle\langle 0|_A \otimes \mathbb{1} + |1\rangle\langle 1|_B \otimes X_B \equiv$



- CNOT generates entanglement:



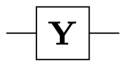
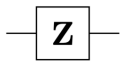
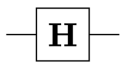
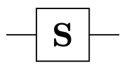
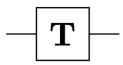
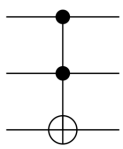
$= CNOT_{(1,2)} (H \otimes \mathbb{1}) |x_1\rangle \otimes |x_2\rangle =$

$= CNOT_{(1,2)} \frac{(|0\rangle + (-1)^{x_1} |1\rangle)}{\sqrt{2}} \otimes |x_2\rangle =$

$= \frac{1}{\sqrt{2}} |0\rangle \otimes |x_2\rangle + \frac{(-1)^{x_1}}{\sqrt{2}} |1\rangle \otimes \overline{|x_2\rangle}$

If $x_1, x_2 = 0 \Rightarrow \frac{|00\rangle + |11\rangle}{\sqrt{2}}$

- OTHER GATES:

Operator	Gate(s)	Matrix
Pauli-X (X)		$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = ie^{-i\frac{\pi}{2}}$
Pauli-Y (Y)		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = ie^{-i\frac{\pi}{2}Z}$
Hadamard (H)		$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{pmatrix}$
π/8 (T)		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

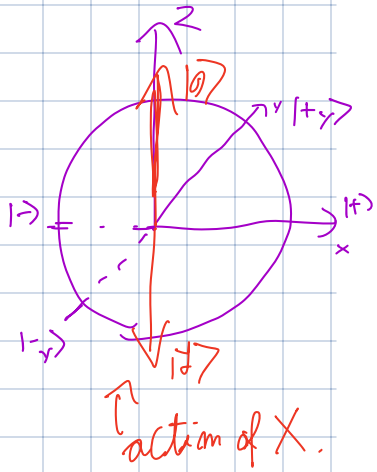
$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = ie^{-i\frac{\pi}{2}X} = \left(\cos\left(\frac{\pi}{2}\right)\mathbb{1} - i\sin\left(\frac{\pi}{2}\right)X\right)i$

$P^2 = \mathbb{1} \Rightarrow e^{-i\frac{\theta}{2}P} = \cos\frac{\theta}{2}\mathbb{1} - i\sin\frac{\theta}{2}P$

$\left(e^A = \sum_{i=1}^{\infty}\frac{A^i}{i!}\right) = R_P(\theta)$

$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{pmatrix} = e^{i\frac{\pi}{4}}\begin{pmatrix} e^{-i\frac{\pi}{4}} & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{pmatrix} = e^{i\frac{\pi}{4}}e^{-i\frac{\pi}{4}Z} = e^{i\frac{\pi}{4}}R_Z\left(\frac{\pi}{2}\right)$

Rotation of π around $\hat{z}$ in the Bloch sphere.



action of X.

- $C\text{-}U_{(A,B)} = |0\rangle\langle 0|_A \otimes \mathbb{1}_B + |1\rangle\langle 1|_A \otimes U_B$

- $HZH = X$, $X = HZH$

$HZH|0\rangle = HZ|+\rangle\ H|-\rangle = |1\rangle = X|0\rangle$
$HZH|1\rangle = X|1\rangle$
$\Rightarrow HZH = X \Rightarrow Z = HXH$ ($H^2 = \mathbb{1}$)

- $(HS^\dagger)^\dagger 2 (HS^\dagger) = Y$ $(SXS^\dagger = Y)$

$(HS^\dagger)^\dagger 2\, HS^\dagger = S H 2 H S^\dagger = S X S^\dagger = Y$

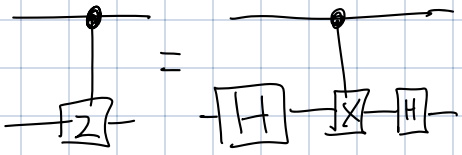
$S X S^\dagger |0\rangle = i|1\rangle = Y|0\rangle$

$S X S^\dagger |1\rangle = -i|0\rangle = Y|1\rangle$

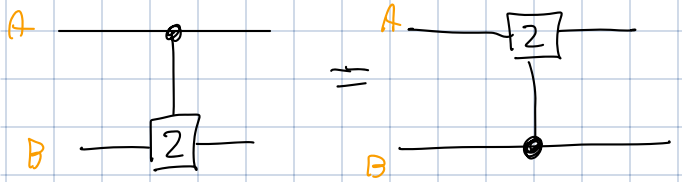
- $C\text{-}2 = |0\rangle\langle 0| \otimes \underbrace{\mathbb{1}}_{HH} + |1\rangle\langle 1| \otimes \underbrace{2}_{HXH}$

$\Rightarrow C\text{-}2 = \mathbb{1}\otimes H \,(C\text{-}NOT)\, \mathbb{1}\otimes H$, $C\text{-}NOT = \mathbb{1}\otimes H \,(C\text{-}2)\, \mathbb{1}\otimes H$

$\uparrow$ $H^2 = \mathbb{1}$, $HXH = 2$



- $C\text{-}2_{(A,B)} = C\text{-}2_{(B,A)}$



- They act equally on computational bases:

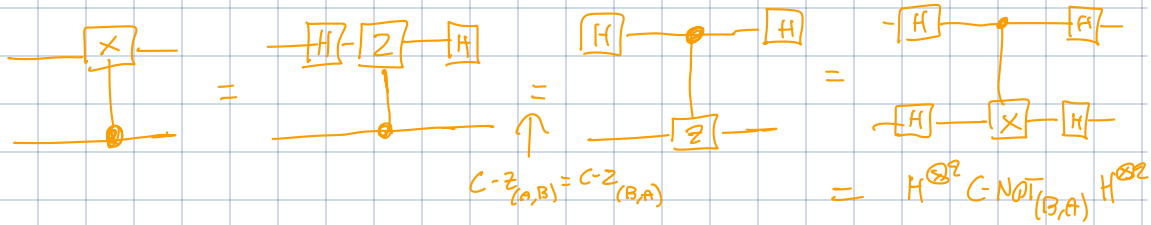
$C\text{-}2_{(A,B)} |x_A\rangle \otimes |x_B\rangle = (-1)^{x_A \cdot x_B} |x_A\rangle \otimes |x_B\rangle$

$= C\text{-}2_{(B,A)} |x_A\rangle \otimes |x_B\rangle$

- $C\text{-}NOT_{(B,A)} = H^{\otimes 2}\, C\text{-}NOT_{(A,B)}\, H^{\otimes 2}$

PROOF:

$C\text{-}NOT_{(B,A)} =$



$= H^{\otimes 2}\, C\text{-}NOT_{(B,A)}\, H^{\otimes 2}$

- SWAP $|i\rangle \otimes |j\rangle := |j\rangle \otimes |i\rangle$ $\quad \forall i, j = 0, 1$

SWAP $|\psi\rangle \otimes |\phi\rangle = |\phi\rangle \otimes |\psi\rangle$

PROOF

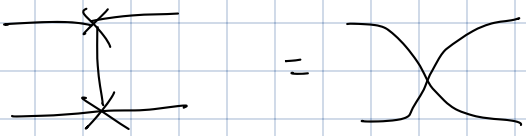
SWAP $|\psi\rangle \otimes |\phi\rangle =$ $\quad \sum_{i,j} \psi_i \phi_j$ SWAP $|i\rangle \otimes |j\rangle = \sum_{i,j} \psi_i \phi_j |j\rangle \otimes |i\rangle =$

$|\psi\rangle = \sum_i \psi_i |i\rangle$

$|\phi\rangle = \sum_j \phi_j |j\rangle$

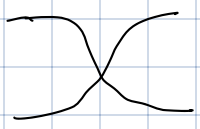
$= |\phi\rangle \otimes |\psi\rangle$

SWAP :=

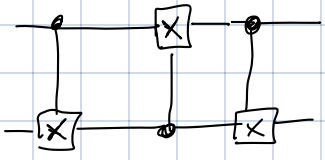


=

$SWAP_{(A,B)} = C\text{-}NOT_{(A,B)}$ $\quad C\text{-}NOT_{(B,A)}$ $\quad C\text{-}NOT_{(A,B)}$

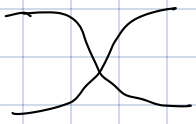


=

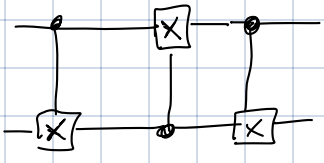


PROOF:

They act equally on $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$



=



=

